

When the Baseline Has No Parameters: Evaluation Pitfalls in Single-Event Industrial Anomaly Detection

Anonymous Submission

Abstract

We report a methodological audit of a widely distributed industrial predictive-maintenance benchmark: vibration and current time series from the hydraulic pump of a fine-blanking press. The benchmark is representative of a large class of public manufacturing datasets—one labelled fault episode, one unlabelled normal recording, acquired on different days. We make four observations that we believe generalise beyond this dataset. (i) The acquisition is burst-sampled: 58% of wall-clock time is gaps, and naive sliding windows span those gaps, inflating the window count by $2\times$ and manufacturing detectable amplitude discontinuities; gap-aware windowing lowers the reference F_1 from 0.857 to 0.818, and the entire loss is in recall. (ii) Window-level F_1 is saturated: a zero-parameter range rule attains $F_1 = 0.857$ against 0.748 for the reference LSTM-autoencoder, so reporting F_1 alone cannot distinguish learned models from trivial ones. (iii) Three commonly used operational metrics degenerate when the positive class is a single event—detection lead time is *unattainable by construction*, event recall is 1.0 for every detector, and a false-alarm rate of 1/hour is unresolvable at the available 0.139 h of normal observation (Poisson 95% upper bound 21.6/h at zero alarms). (iv) The current channel separates the two recordings perfectly ($F_1 = 1.000$ from two features alone), but its mains carrier occupies 95.9% of normal AC power versus 10.4% in the fault recording—physically impossible for a running motor, and therefore an acquisition artefact rather than a fault signature. We adopt a pre-declared exclusion rule, quantify its cost at $\Delta F_1 = 0.072$, and show that a vibration-only detector still reaches $F_1 = 0.928$ (95% CI 0.852–0.984, zero false positives, zero learned parameters). Finally, we localise 91.1% of all false alarms to a single reversible 300 s low-load excursion inside the *normal* recording, which we argue is the genuinely informative finding the benchmark affords.

1 Introduction

Public benchmarks for industrial anomaly detection are typically small, weakly labelled, and assembled from a handful of recording sessions. A recurring pattern—visible in the dataset we study and in many of its peers—is that the “normal” and “anomalous” classes correspond to *different recording sessions*. Any difference between sessions, whether mechanical or instrumental, is then perfectly confounded with the label.

This is not a hypothetical concern. Wu & Keogh [2021] showed that several widely used time-series anomaly benchmarks are solvable by one-line rules; Kim et al. [2022] showed that the point-adjustment protocol common in the literature inflates F_1 to the point where random scores outperform published models. Our contribution is complementary and concrete: we take a single benchmark that is actively used in a national competition, and we trace, with measured numbers, exactly which evaluation decisions change the conclusion and by how much.

We organise the paper around the decisions an analyst must make, in the order they arise: how to window burst-sampled data (§2), which metrics remain identifiable when the positive class is one event (§3), how to decide whether a highly discriminative channel is signal or artefact (§4), and only then which detector to use (§5–§6). Our view is that the first three decisions dominate the fourth, and that reporting practice should reflect that ordering.

Contributions.

- We identify a burst-sampling acquisition structure and show that gap-unaware windowing discards the distinction between adjacent and non-adjacent samples, inflating recall through artefacts (§2).
- We show window F_1 on this benchmark is saturated by a zero-parameter rule, and propose a two-tier baseline protocol that preserves both an honest improvement margin and a strict reference (§3).
- We give a falsifiable, pre-declared exclusion rule for session-confounded channels and report the exclusion cost rather than hiding it (§4).
- We localise the dominant false-alarm mode to a reversible operating-mode excursion inside the normal class, and show it is the mechanism behind the apparent robustness of amplitude features (§6.4).

2 Dataset and Acquisition Structure

2.1 The benchmark

The dataset provides two CSV files sampled at a nominal 10 Hz with three analogue channels—upper motor vibration (AI0), lower motor vibration (AI1), and motor current (AI2)—plus a constant `Equipment_state` flag. The normal file contains 20,000 rows recorded on 2022-07-12 (77 min wall

Table 1: Effect of windowing on the zero-parameter range rule. All rows use identical splits and thresholds; only the windowing differs. The recall-only loss indicates that gap-straddling windows were being detected for instrumental, not mechanical, reasons.

Windowing	$N_{\text{norm}}/N_{\text{fault}}$	P	R	F_1
naive, offset 100	4000 / 180	1.000	0.750	0.857
naive, no reserve	4880 / 480	1.000	0.704	0.826
gap-aware, offset 100	2433 / 239	1.000	0.649	0.787
gap-aware	2489 / 276	1.000	0.692	0.818

clock); the fault file contains 600 rows recorded on 2022-07-17 (2.8 min). Every row of the normal file is labelled 0 and every row of the fault file is labelled 1: the label is a *file* attribute, not a row attribute. There is therefore exactly one positive event.

2.2 Burst sampling

Consecutive-sample intervals are not uniform. Taking a gap threshold of 0.5 s, the normal file decomposes into 599 contiguous runs and the fault file into 21. Run lengths are capped at exactly 50 samples (5 s) in both files, with 33.7% of normal runs attaining the cap. Gaps account for 58.0% of normal wall-clock time and 65.0% of fault wall-clock time. The inter-burst start interval is tightly distributed (median 7.96 s) while burst length is not, which is the signature of a periodic publish-and-drop transport layer rather than a process cycle.

This matters for windowing. With $L = 20$ samples, only 9,978 of the 19,981 naive stride-1 windows lie entirely inside a run (49.9% survival); at $L = 10$ survival is 74.4% and at $L = 5$ it is 88.3%. A naive window that straddles a gap concatenates signal segments separated by up to 16.6 s and so contains a discontinuity of purely instrumental origin.

Two consequences follow. First, nominal window duration is wrong: a naive 20-sample window spans a median 3.35 s (max 24.8 s) of wall clock, not 2 s; gap-aware windows span exactly 1.9 s. Second—and this is the quantitative point—removing straddling windows lowers the reference zero-parameter detector from $F_1 = 0.857$ to 0.818, and the drop is *entirely* in recall ($0.750 \rightarrow 0.692$) while precision and FPR stay at 1.000 and 0.000. The discarded windows were therefore not informative detections but gap-induced amplitude jumps.

2.3 Spectral structure

The reference analysis distributed with the dataset characterises normal vibration as uncorrelated. Inspecting lag-1 autocorrelation alone supports this ($r_1 = -0.011$ for A10), but it is an artefact of the chosen lag. Burst-averaged periodograms (50 samples, Hann window, 0.2 Hz resolution, uniform expectation 3.8% per bin) place 25.0% of A10 power in a single 3.6 Hz bin and 38.0% of A11 power in a single 1.8 Hz bin, in a stable 2:1 ratio present in every burst. A Ljung–Box test [Ljung & Box, 1978] over lags 1–8 rejects whiteness in 275 of 276 normal A10 bursts. Normal operation is thus a narrowband line spectrum, and the fault condition is the *collapse* of that

line structure into broadband low-frequency content.

We note that at 10 Hz the Nyquist limit is 5 Hz, whereas a gear-type hydraulic pump driven by a four-pole induction motor has a gear-mesh frequency of roughly 300 Hz. Bearing and gear fault frequencies are therefore not recoverable from the vibration channels under any transform, which rules out the envelope-spectrum family of methods regardless of modelling choice. We make this explicit because the conclusion is insensitive to the pole-count assumption: 2-pole (60 Hz) and 6-pole (20 Hz) shafts both exceed Nyquist.

3 Which Metrics Survive One Event?

3.1 Degenerate operational metrics

Operational anomaly-detection practice favours event-level metrics over sample-level ones [Hundman et al., 2018, Garg et al., 2021]. On this benchmark three such metrics are not merely noisy but *unidentifiable*.

Detection lead time is unattainable. Lead time is defined as $t_0 - t_{\text{alarm}}$ where t_0 is fault onset. The fault file is labelled positive from its first row, so t_0 can only be taken as that row and no alarm can precede it. A window of $L = 20$ requires 1.9 s to fill and a $k = 5$ debounce a further 0.5 s, bounding lead time above by -2.4 s. The target region is empty. We replace lead time by *detection delay*.

Detection delay is dominated by acquisition. Every detector we evaluate reports an identical delay of 8.78 s. The cause is structural: the first burst of the fault file is 4 samples long and admits no window, so the first evaluable window begins 8.78 s after t_0 . Detection delay therefore measures the data, not the model.

Event recall is constant. With a single positive event, every detector that fires at all scores 1.0. The Clopper–Pearson interval is $[0.025, 1.0]$.

False-alarm rate is unresolvable. The normal test segment provides 0.139 h of data time. Observing zero alarms gives a Poisson 95% upper bound of 21.6 alarms/h; one alarm gives 34.2/h. A target of ≤ 1 alarm/h would require roughly 3 h of alarm-free normal observation. Treating “0.0/h” as a hard admission criterion therefore filters candidates on noise. We report false-positive *counts* and use them only for ranking.

3.2 What remains

After the above, the identifiable axes are the bootstrap confidence interval of window F_1 , the false-positive count, and model complexity. Because stride-1 windows overlap by 19/20 samples, we resample at the burst level: all confidence intervals and paired comparisons use a block bootstrap over bursts ($B = 1000$), which is the smallest unit that is plausibly exchangeable.

3.3 Two-tier baselines

A zero-parameter rule that flags any window leaving the per-channel training range attains $F_1 = 0.857$ under the reference protocol, against 0.748 for the LSTM-autoencoder [Malhotra et al., 2016] distributed with the dataset. Window F_1 is thus saturated near 0.85, and a deep model scoring 0.80 is not “good” but worse than no model.

However, promoting this rule to *the* baseline is also wrong, for a reason specific to the data: the extrema of all three channels occur before row 15,000, so the test segment lies strictly inside the training range and the rule *cannot* produce a false positive. Its FPR = 0 is an identity, not a measurement. We therefore report two tiers—the conventional reference model and the strict zero-parameter rule—and we verify, for every range-based detector we report, that the training range is in fact exceedable on test data (it is, for 2.6–3.8% of windows).

4 Session Confound and a Pre-declared Exclusion Rule

4.1 The current channel separates the sessions, not the classes

The lag-1 autocorrelation of A12 is 0.927 in the normal file with a minimum of 0.906, and at most 0.836 in the fault file: the two classes are perfectly separated by one scalar. Taken at face value this is the strongest feature in the dataset. Three observations argue it is an artefact.

First, normal A12 is a pure aliased tone. A single sinusoid fits each burst with median $R^2 = 0.996$ at an apparent 0.600 Hz, and $\cos(2\pi \cdot 0.6/10) = 0.9298$ reproduces the observed autocorrelation. This identifies a 60 Hz mains carrier folded by a sampling clock in error by 1.0%. Second, the statistic is a function of carrier *frequency*, which load cannot change, so its invariance across operating conditions carries no information about whether it tracks machine health. Third, and decisively, the share of AC power in the carrier band falls from 0.959 (normal, range [0.947, 0.973]) to 0.104 (fault, [0.046, 0.183]) with no overlap. For a running induction motor the mains fundamental cannot vanish; a change of this magnitude indicates a different acquisition path.

We also note a file-level fingerprint independent of any channel: decimal precision differs between the files (median 6 digits with 0.0% exceeding seven in the normal file, versus median 8 and 64.8% in the fault file, including float64 serialisation residue such as -44.1074420000000006). The two files were exported by different pipelines. The confound is therefore present at the file level and is not specific to any one variable.

4.2 The rule

Because a single fault session cannot falsify an instrumental explanation, we decline to adjudicate it and instead pre-declare an exclusion rule:

D1. A feature that is invariant (coefficient of variation $< 1\%$) across all observed normal operating variation, yet differs between acquisition sessions, is excluded from the final model, because a single session cannot falsify an instrumental explanation.

D1 is a procedural rule, not a physical verdict. It applies to the whole A12 channel, including the amplitude features A12_std and A12_ptp, which are functions of the same carrier. We regard partial exclusion—banning the autocorrelation while retaining the amplitude—as indefensible, and we report the cost of full exclusion in §6.

5 Features and Detectors

5.1 Feature sets

All features are computed on gap-aware windows of $L = 20$. We evaluate four D1-compliant sets, each built to test one hypothesis about what distinguishes the fault, plus two D1-violating sets retained only for sensitivity analysis.

- S0_AMP4: $\{\sigma, p2p\}$ for A10, A11. Amplitude only.
- S1_VIB: S0 + lagged correlations at lags 1, 2 + $\rho(\text{A10}, \text{A11})$ + amplitude ratio + interaction.
- S2_SHAPE: amplitude + kurtosis, crest factor + ratio + interaction.
- S3_SPEC: amplitude + band power at 1.5–2.5 and 3.0–4.0 Hz + ratio + interaction.

Two constructions deserve comment. The *amplitude ratio* $\sigma(\text{A10})/\sigma(\text{A11})$ is invariant to per-channel gain and polarity, which matters because the dataset documents no sensor axis, mounting angle, or unit; consequently no absolute vibration-severity standard can be applied, and all thresholds must be relative to the data itself. The *interaction* term $-\rho \cdot \sigma(\text{A10})$ encodes a finding from §6.4: antiphase motion alone is not diagnostic because it also occurs in normal operation; antiphase *together with* increased amplitude is.

5.2 Detectors

All detectors are one-class and are fitted on normal training windows only: the range rule; Mahalanobis distance and a one-sided variant that accumulates only positive deviations; principal-component T^2 and squared prediction error [Jackson & Mudholkar, 1979, MacGregor & Kourti, 1995]; Isolation Forest [Liu et al., 2008]; Local Outlier Factor in novelty mode [Breunig et al., 2000]; and a one-class SVM [Schölkopf et al., 2001]. Thresholds are the $q_{99.9}$ quantile of the score on the normal validation segment, so no fault label enters threshold selection. Scores are converted to p -values by split conformal prediction [Vovk et al., 2005, Laxhammar & Falkman, 2015], which requires only normal data; we use the uniformity of normal p -values as a calibration diagnostic rather than claiming a coverage guarantee, since exchangeability is violated by the excursion of §6.4.

Table 2: D1-compliant detectors, ranked by the lower bound of the bootstrap F_1 interval. FP is the count of false positives among 2489 normal test windows; $|\theta|$ is the number of learned parameters.

Features	Detector	F_1	95% CI	FP	$ \theta $
S1_VIB	Range rule	0.928	[.852,.984]	0	0
S3_SPEC	PCA-SPE	0.917	[.849,.971]	10	55
S3_SPEC	Mahalanobis	0.905	[.829,.970]	11	65
S2_SHAPE	OCSVM	0.903	[.817,.976]	17	95
S2_SHAPE	LOF	0.896	[.808,.973]	4	0
S2_SHAPE	Mahalanobis ^{1S}	0.888	[.794,.971]	22	65
S2_SHAPE	Range rule	0.877	[.786,.955]	6	0
S1_VIB	PCA- T^2	0.863	[.791,.927]	60	72
S0_AMP4	OCSVM	0.866	[.773,.949]	7	72

6 Results

6.1 Protocol

Train on normal rows [0, 12000) (6,021 windows), select thresholds on normal rows [12000, 15000) (1,468 windows), and test on normal rows [15000, 20000) (2,489 windows) together with all 276 fault windows. Standardisation is fitted on the training segment only and is omitted for the range rule and Isolation Forest, which are invariant to monotone per-feature transforms. Confidence intervals are burst-level block bootstraps with $B = 1000$.

6.2 Main result

Table 2 gives the D1-compliant leaderboard. The best detector is a zero-parameter range rule on S1_VIB, at $F_1 = 0.928$ with perfect precision and no false positives. Against the strict baseline of §3 ($F_1 = 0.818$) the paired block bootstrap gives $\Delta F_1 \in [+0.008, +0.202]$ with $P(\Delta \leq 0) = 0.019$; against a Mahalanobis detector on amplitude features only (0.828) it gives $[+0.055, +0.159]$, $P(\Delta \leq 0) = 0.000$.

We verify that this result is not the identity of §3: in the S1_VIB feature space, 95 of 2489 normal test windows (3.82%) do exceed the training range on at least one coordinate, and are nonetheless scored below threshold.

The second-ranked detector, PCA-SPE on spectral features, is *not distinguishable* from the first: $\Delta F_1 \in [-0.022, +0.043]$, $P(\Delta \leq 0) = 0.265$. We consider this the most useful single number in the study. Rather than break the tie on point estimates, we assign the two models different roles: the zero-parameter rule is the *reference* that any more complex model must beat significantly, and the PCA model is the *deployment* candidate, because its residual decomposition yields contribution plots that attribute an alarm to individual variables—the standard diagnostic artefact in multivariate process control.

6.3 Cost of the exclusion rule

Table 3 quantifies what D1 costs. Two current-derived features suffice for perfect separation. We report this prominently

Table 3: Sensitivity to D1. Excluding the current channel costs $\Delta F_1 = 0.072$. Two current-derived features alone separate the classes perfectly, which we read as a measure of the confound rather than of performance.

Features	Detector	F_1	D1
AI2 only (2 feat.)	PCA-SPE	1.000	violates
AI2 only (2 feat.)	Mahalanobis	1.000	violates
amp. + AI2	LOF	0.984	violates
amp. + AI2	OCSVM	0.964	violates
S1_VIB (adopted)	Range rule	0.928	complies

Table 4: False alarms are not uniformly distributed over the normal test segment. Rows are 1000-sample bins; statistics are bin means. The detector is the worst-performing D1-compliant candidate (191 false positives), chosen to make the concentration visible.

Rows	n	FP	rate	σ_{AI1}	ρ_{01}
15–16k	501	13	2.6%	0.089	+0.197
16–17k	499	64	12.8%	0.043	−0.198
17–18k	504	56	11.1%	0.042	−0.248
18–19k	502	54	10.8%	0.058	−0.093
19–20k	483	4	0.8%	0.103	+0.316

precisely because it is the exclusion’s justification: given independent evidence that the current acquisition path differed between sessions, perfect separation measures the confound, not the detector. The cost of declining it is $\Delta F_1 = 0.072$.

6.4 Where the false alarms are

Table 4 shows that 91.1% of false positives fall in rows 16,000–19,000, roughly 300 s. In that interval vibration amplitude drops by 53% and the upper–lower correlation reverses sign, yet the label remains normal; by row 19,000 both statistics return to training levels (0.103 and +0.316 against 0.115 and +0.306). This is a reversible second operating mode, not a regime change, and we were able to confirm that it co-varies with a load proxy.

Two implications follow. First, sign reversal of inter-sensor correlation is *not* a fault signature: it occurs in labelled-normal data. What distinguishes the fault is reversal *accompanied by rising* amplitude, whereas the normal excursion reverses while amplitude *falls*—which is why we include the interaction feature of §5. Second, and less comfortably, the zero false positives of our adopted detector are a consequence of direction rather than robustness: normal operating variation pushes amplitude *inside* the training support while the fault pushes it outside. A normal mode that raised amplitude—a heavier blanking job, a higher pressure setting—would produce false alarms. No such mode is present in these data, and we state this as a limitation rather than discovering it later.

7 Discussion

Ordering of decisions. Across every comparison we ran, the choice of windowing ($\Delta F_1 = 0.039$), the choice of whether

to admit the confounded channel (0.072), and the choice of baseline (the difference between claiming +0.18 and claiming +0.11 over the reference model) each moved the headline number by more than the spread across all eight detectors on a fixed feature set. Model selection was the least consequential decision we made. We suspect this ordering is common in small industrial benchmarks and is under-reported because the preprocessing decisions are rarely ablated.

On zero-parameter winners. That a range rule tops Table 2 is not an argument against learning; it is a statement about the identifiability of this benchmark. With 276 positive windows drawn from one event, the effective positive sample size is one, and the bootstrap intervals in Table 2 overlap almost completely. The honest reading is that this dataset can rank preprocessing choices but cannot rank detectors.

Calibration versus coverage. We deliberately do not claim a conformal coverage guarantee. Exchangeability fails across the excursion of \$6.4, and in feature sets that include correlation statistics we measured empirical false-positive rates exceeding the nominal α by up to $84\times$. We report the Kolmogorov–Smirnov distance of the normal p -value distribution from uniform as a diagnostic instead; it is smallest (0.046–0.105) exactly for the amplitude and spectral feature sets that are insensitive to the excursion.

Limitations. The study concerns one dataset with one fault event, and all statements about the fault mechanism are necessarily weak. We make no claim about which failure mode produced the recording; the available bandwidth (≤ 5 Hz) excludes the diagnostic frequencies that would be required. Our evaluation protocol also inherits the dataset’s single-session structure: a model that passes D1 may still be exploiting session differences that we failed to identify.

8 Conclusion

On a representative industrial predictive-maintenance benchmark, we find that three evaluation decisions—gap-aware windowing, baseline selection, and the treatment of a session-confounded channel—each affect the reported F_1 more than the choice among eight standard one-class detectors, and that three commonly reported operational metrics are unidentifiable when the positive class is a single event. We recommend that work on such benchmarks report the windowing survival rate, verify that range-based baselines can in principle produce false positives, state the effective number of positive events alongside any interval estimate, and quantify the cost of excluding any channel suspected of acquisition confounding. Applying this protocol, a vibration-only detector with no learned parameters attains $F_1 = 0.928$ with zero false positives, and 91.1% of all false alarms in the study localise to a single reversible load excursion that the benchmark labels normal—which we regard as the finding the data actually supports.

Reproducibility. The full pipeline runs end to end from a single entry point and emits the comparison table, the false-alarm decomposition, and per-window predictions. All randomised components are seeded; the range rule, Mahalanobis, and PCA detectors are deterministic. Repeated runs are byte-identical.

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